

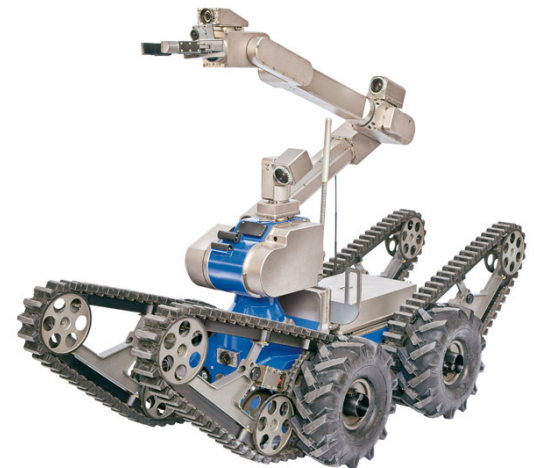


1st WORKSHOP FTEC-2015 SPANISH TRAINEESHIP PROGRAMME

Robotic projects low level control development



Alejandro Diaz Fontalva
Advisor: *Mario Di Castro*
Section: *EN-STI-ECE*
Ref. Number: *EN2734*



ENGINEERING
DEPARTMENT

Ciemat
Centro de Investigaciones
Energéticas, Medioambientales
y Tecnológicas

Train Inspection Monorail (TIM)

Goal

- ❖ Development of Unmanned tasks along LHC

Features

- Attached to ceiling
- 5 Wagons
- 3G UMTS / Wi-fi
- LabVIEW GUI



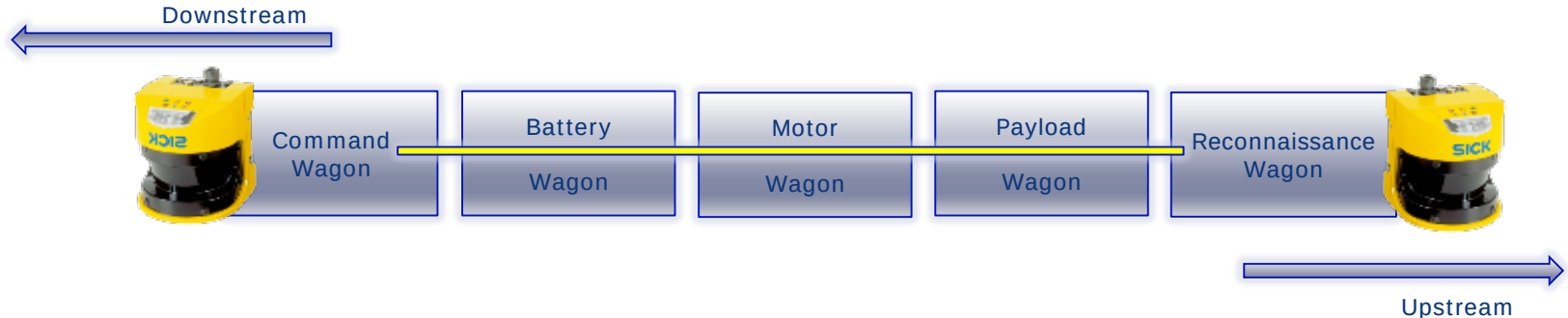
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Train Inspection Monorail (TIM)

MainTimPcClient_V11.lvi

File Edit View Project Operate Tools Window Help

TIM Train Inspection Monorail

Operator: timUSER Refresh Cameras

Operation Train Feedback Movement RP Survey Measurements Service Expert Guru

Head Camera Tail Camera RP Arm Camera Charging Arm Camera Thermal Camera

Train Status
Charging

Train Moving
Stop Train
STOP

Communication
ON

Battery Capacity [%]
100

Odometer [m]
49334.5

Laser Scanner
Scanner Status: ON
Front Scanner Warning Field
Back Scanner Warning Field

Feedback
Obstacles in the area - No RP arm extraction possible
Speed is decreased to 0.3 m/s
Obstacle in this area: ChicanesBarcode Reader Error

Check Obstacle Database

Open Map

Train Position [m] DCUM
13370

LHC

AXIS 213 PTZ Network Camera
Live View | Setup | Help

Video format: Motion JPEG Source: Camera front Thu Apr 16 2015 16:04:57

Up TILT Down

PAN Left Right ZOOM Wide Tele Ctrl panel FOCUS Near Far IRIS Close Open

Train Feedback

Train Status
Charging

Speed Indicator [m/s]
0

Train Position [m] DCUM
13370

Stop Moving
STOP

Train Moving

Communication
ON

EMERGENCY STOP

Train Inspection Monorail (TIM)

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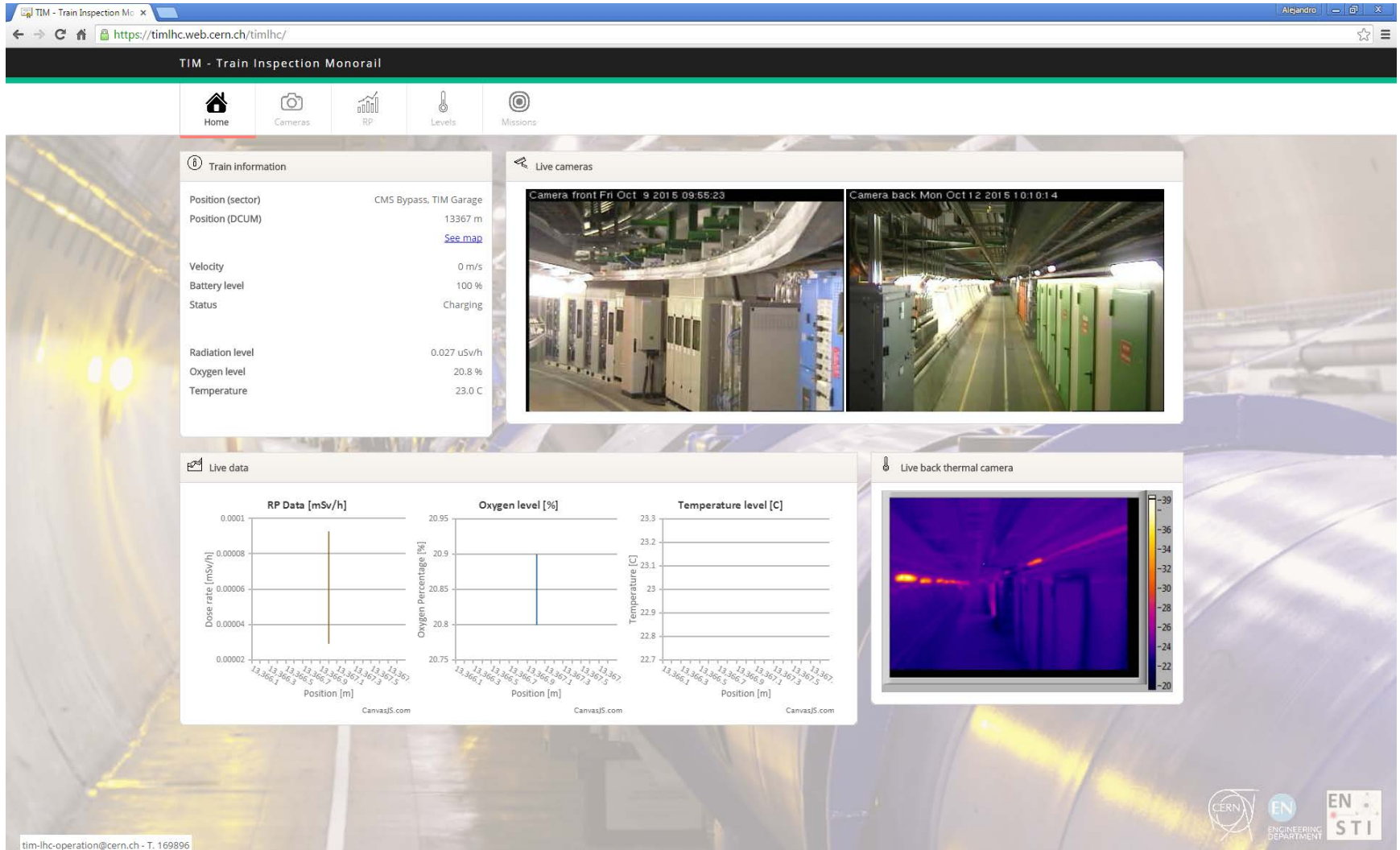
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Main Tasks

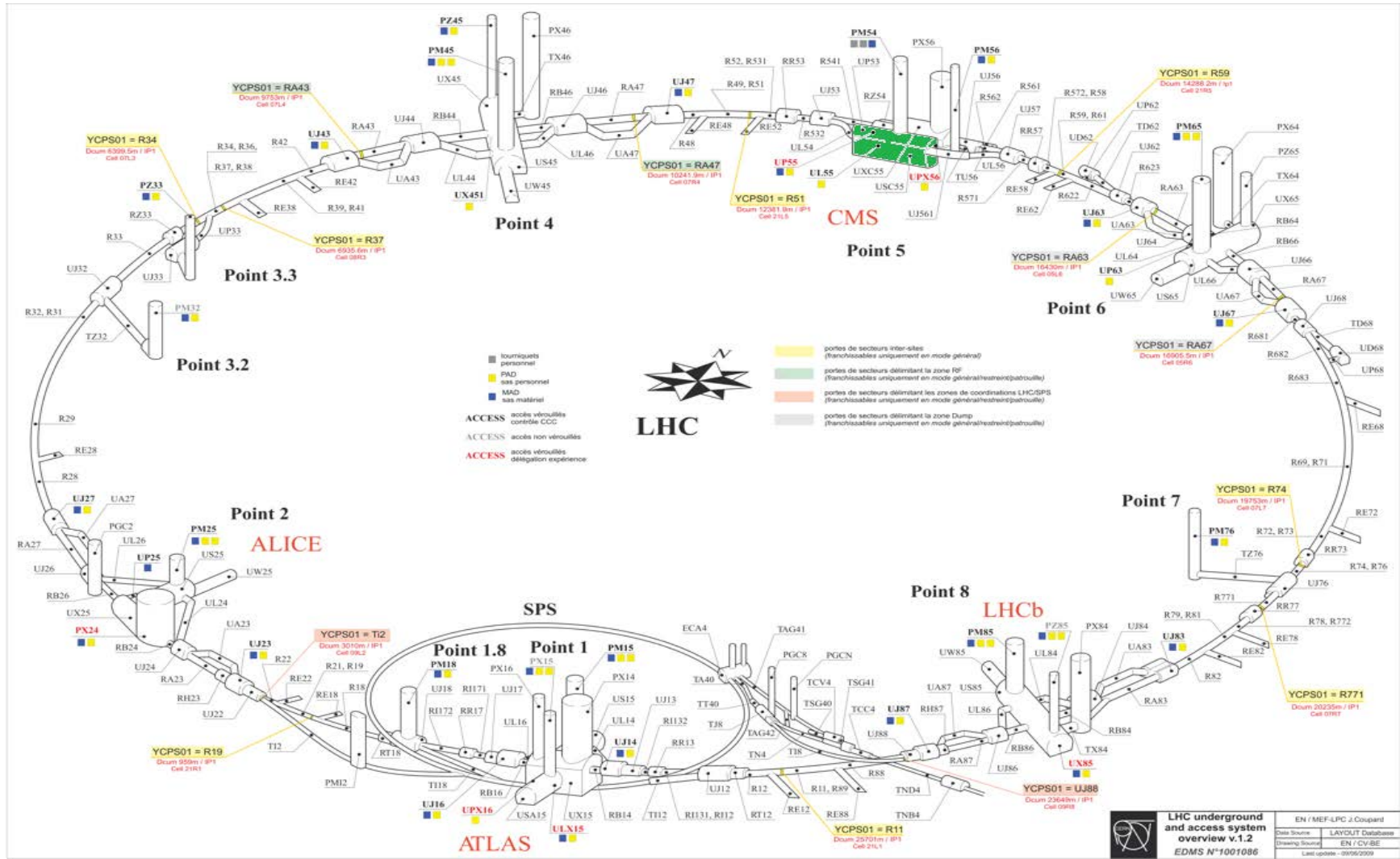
- ❖ Checking of collimators displacements.
- ❖ Measuring of RP dose and Temperature, TIM position, images, etc.



Monitoring and Measurements Treatment



Monitoring and Measurements Treatment

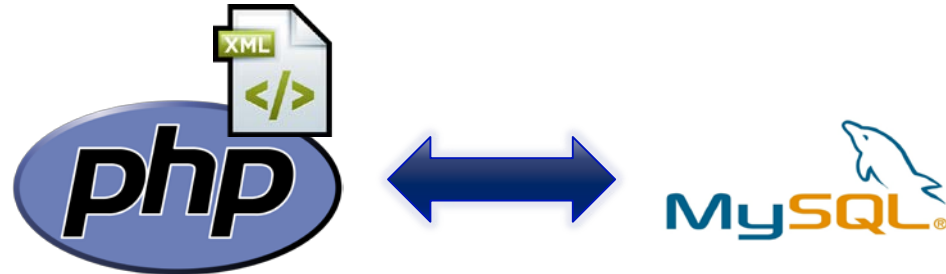


Monitoring and Measurements Treatment

Goals

- ❖ Data Management
- ❖ Graphical information
- ❖ Show images
- ❖ Check TIM position

Contribution



Monitoring and Measurements Treatment

Management options

- Insert new Missions in DB
- Queries to the DB
- Show list of the inserted Missions

Monitoring and Measurements Treatment

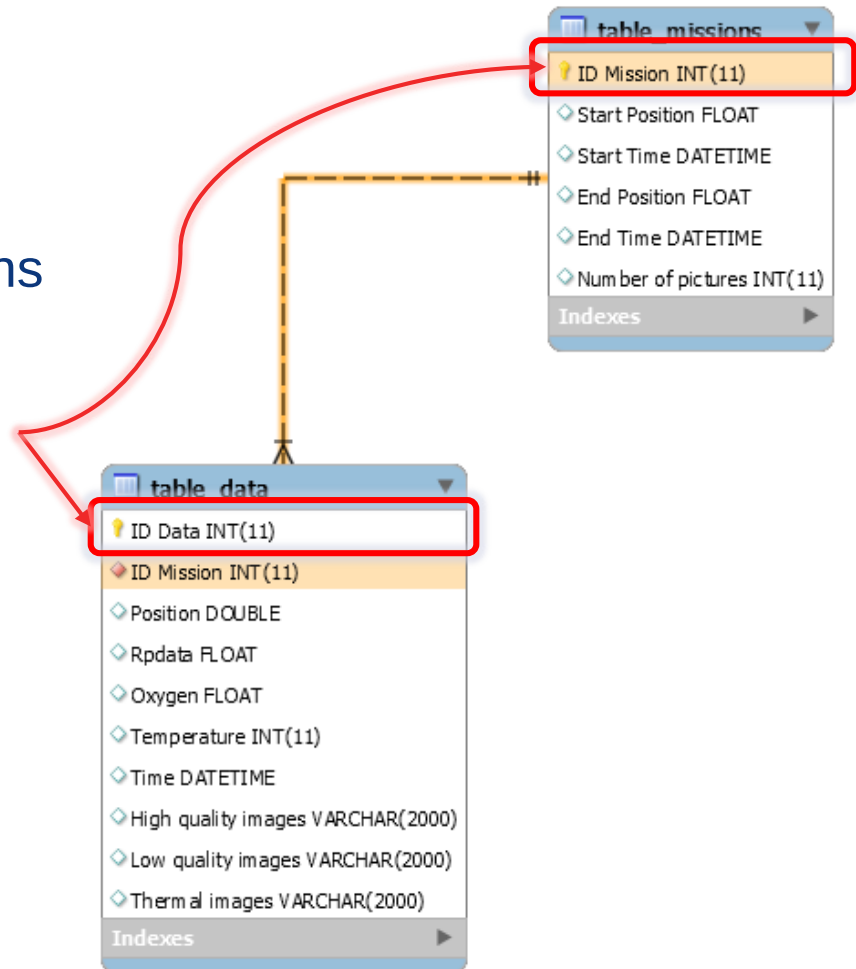
Management options

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2 Tables

- **Table Missions**
- **Table Data**

**PRIMARY
KEYS**



Monitoring and Measurements Treatment

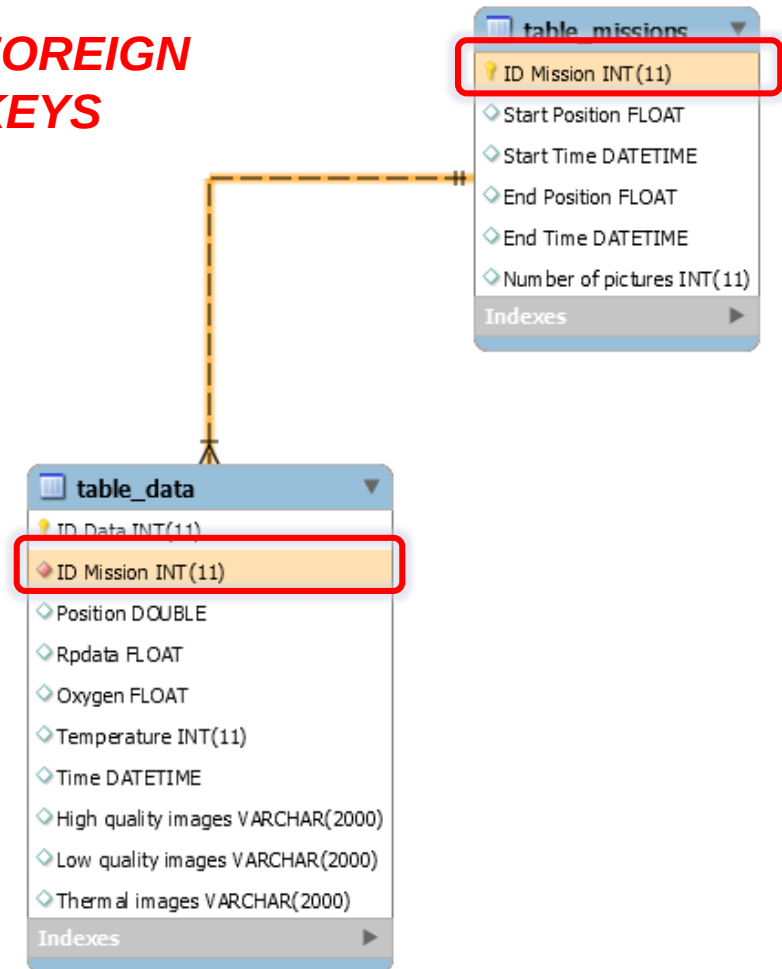
Management options

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**FOREIGN
KEYS**

2 Tables

- **Table Missions**
- **Table Data**



Monitoring and Measurements Treatment

https://timlhc.web.cern.ch x

← → ↻ 🏠 🔒 https://timlhc.web.cern.ch/timlhc/model/TIM_DB_rw.php ☆ ☰

Connected successfully
 Are going to be updated the Missions
 The Mission Table and the Data Table were created in the TIM_DB Database
 The Mission are going to be updated in the DB

Table 1 Missions	ID Mission	Start Position	Start time	End Position	End time	Number of pictures
	1	13174.96	2015-07-09 12:20:33	12824.92	2015-07-09 12:35:46	157

Table 2 Data	ID Data	ID Mission	Position	Rpdata	Oxygen	Temperature	Time	Highq images	Lowq images	Thermal images
	1	1	13174.96	0.366	21.3	25	2015-07-09-12:20:36			
	2	1	13172.87	0.24	21.3	25	2015-07-09-12:20:53			
	3	1	13170.84	0.208	21.3	25	2015-07-09-12:20:58			
	4	1	13168.74	0.145	21.3	25	2015-07-09-12:21:03			

Monitoring and Measurements Treatment

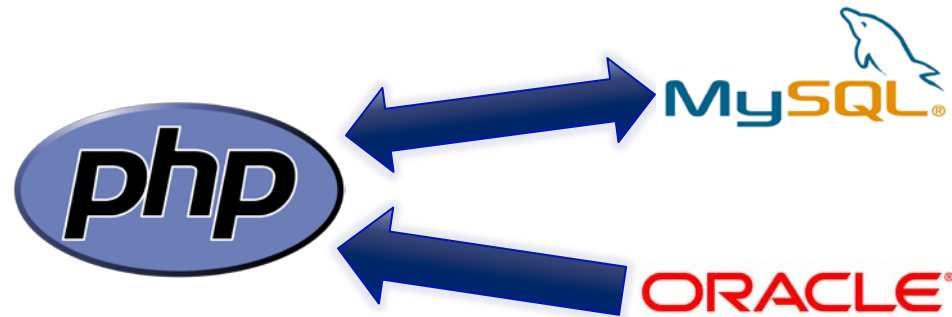
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New Requirements



Monitoring and Measurements Treatment

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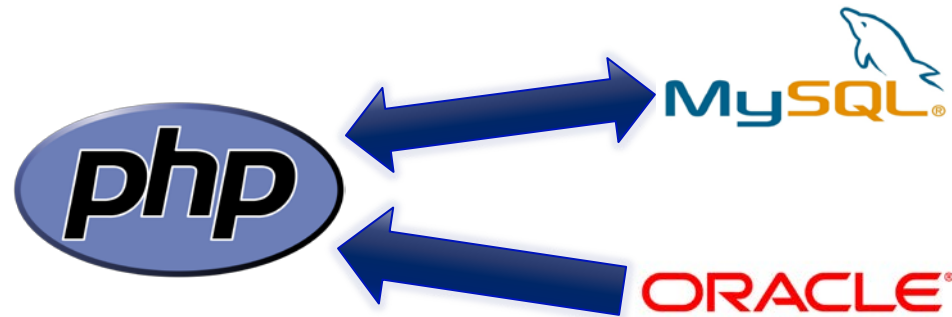
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New Requirements

Challenges

- **ERROR** Uploading Images



Monitoring and Measurements Treatment

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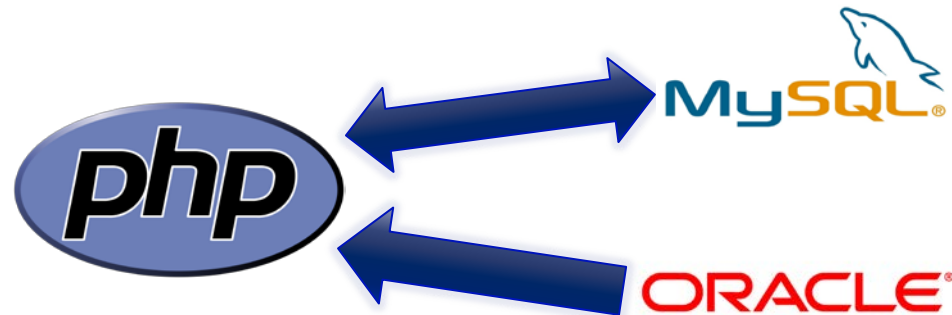
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New Requirements

Challenges

- **ERROR** Uploading Images

Status: Operational



Telemanipulation force feedback

Goal

- ❖ Force feedback in telemanipulated robotic arms

Telemanipulation force feedback

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- ❖ Force feedback in telemanipulated robotic arms

Starting idea

- Bracelet user + Sensors in gripper + Wireless Comm.



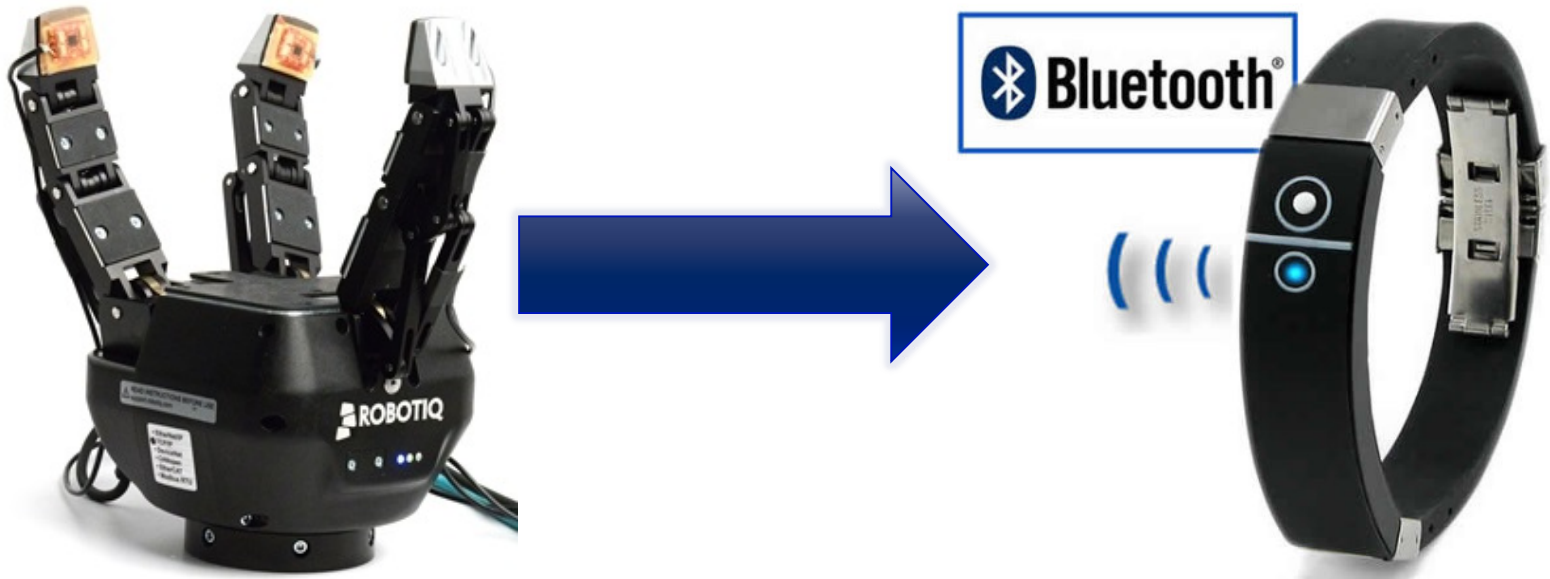
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Telemanipulation force feedback

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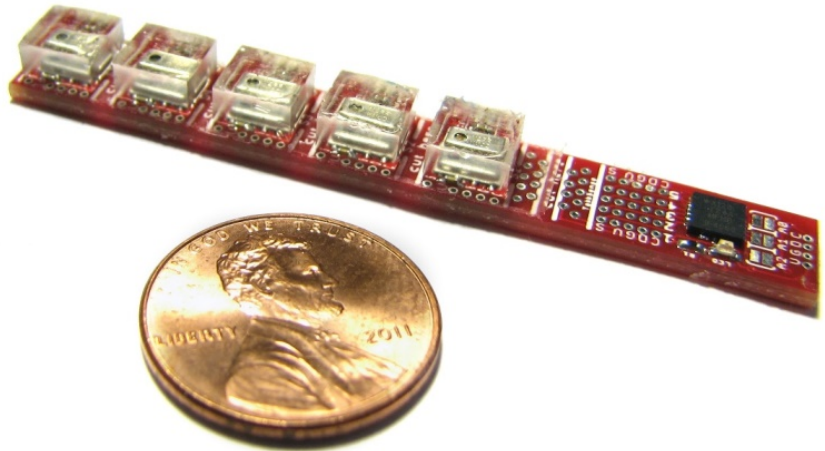
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First tests

- TakkTile sensors



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Challenges

- Change TakkTile → Flexiforce



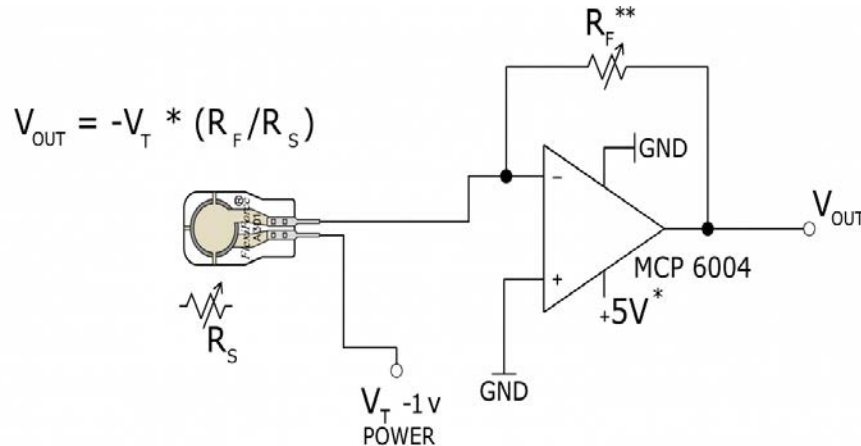
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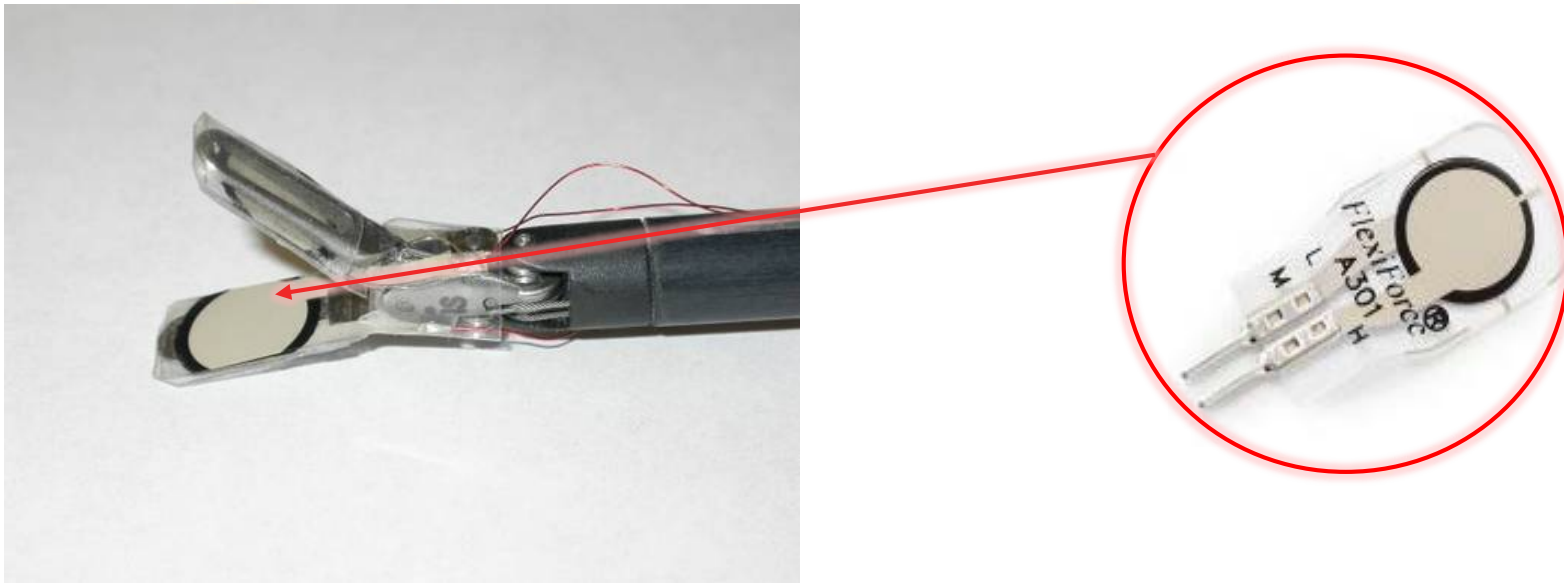
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Status: R&D

Future Guideline

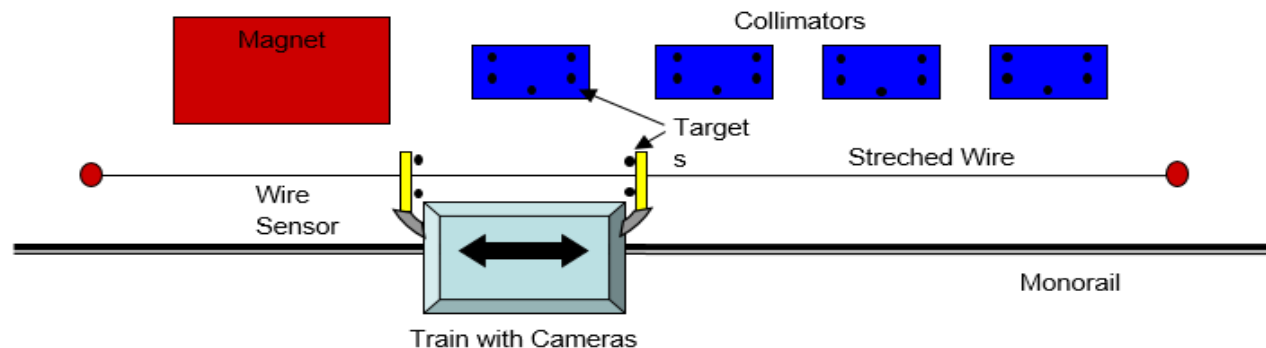
- Continue working on DB and Force Feedback Projects

Future Guideline

➤ Continue working on DB and Force Feedback Projects

➤ 2 New Projects:

❖ Integration of Survey Wagon in TIM.



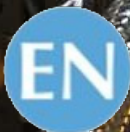
Future Guideline

- Continue working on DB and Force Feedback Projects
- 2 New Projects:
 - ❖ Integration of Survey Wagon in TIM.
 - ❖ Reverse Engineering of Robotic arm.





THANK YOU FOR
YOUR ATTENTION



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